Lecture 29

STAT 109: Introductory Biostatistics — Randomized controlled trials and inference for experiments

Lecture 29

Random assignment and what it buys you

This lecture focuses on **randomized controlled trials (RCTs)**. **Random assignment** of the group membership is the design feature that **replaces** the main threat to internal validity, confounding.

Learning objectives

- Describe an **RCT** for a response and explanatory variable of interest and explain how random assignment could be carried out.
- List **two** primary explanations for an observed difference in an RCT (**treatment effect**, **chance**)
- Distinguish **random assignment** and **random sampling** and say what each
- For a **numerical** outcome and two groups, compute **Lecture 25** descriptives, plot **boxplots / histograms**, and run **Welch's** two-sample t-test (`t.test(..., var.equal = FALSE)`) after checking **conditions**.

RCT definition

A Randomized Controlled Trial (RCT) is a study design where the researcher randomly assigns people/units to one of two or more groups. This assigned group membership becames the explanatory variable and a response variable of interest is measured for each unit.

Internal validity in an RCT

Internal validity = how well the study rules out alternative explanations for an observed difference between the groups within the study.

Under ideal conditions, random assignment balances both measured and unmeasured baseline confounders on average.

Numerical outcome: sleep condition and improvement

Research question. Does **sleep deprivation** change **improvement** in reaction time from a baseline test to a test three days later?

Design (for class). Suppose students were **randomly assigned** to **deprived** vs **unrestricted** sleep groups before the follow-up test of reaction times.

Hypotheses (population means). Let μ_{dep} and μ_{unr} be mean **improvement** in the two groups.

- H_0: $\mu_{\text{dep}} = \mu_{\text{unr}}$
- H_a: $\mu_{\text{dep}} \neq \mu_{\text{unr}}$

Data ($n = 21$)

sleep condition	improvement
unrestricted	-14.7
unrestricted	14.5
deprived	2.2
deprived	21.3
unrestricted	11.5
deprived	13.7
deprived	9.6
deprived	22.4
deprived	21.1
deprived	7.2
deprived	10.0
unrestricted	25.2
unrestricted	-11.2
unrestricted	7.0
unrestricted	14.7
deprived	4.5
unrestricted	45.6
unrestricted	15.6
unrestricted	18.8
unrestricted	12.1
unrestricted	10.3

```r
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

sleep_rct <- tibble(
  sleepcondition = c(
    "unrestricted", "unrestricted", "deprived", "deprived", "unrestricted",
    "deprived", "deprived", "deprived", "deprived", "deprived",
    "deprived", "unrestricted", "unrestricted", "unrestricted", "unrestricted",
    "deprived", "unrestricted", "unrestricted", "unrestricted", "unrestricted",
    "unrestricted"
  ),
  improvement = c(
    -14.7, 14.5, 2.2, 21.3, 11.5, 13.7, 9.6, 22.4, 21.1, 7.2, 10.0,
    25.2, -11.2, 7.0, 14.7, 4.5, 45.6, 15.6, 18.8, 12.1, 10.3
  )
) |>
  mutate(
    sleepcondition = factor(sleepcondition, levels = c("deprived", "unrestricted"))
  )
```

Descriptive statistics (Lecture 25)

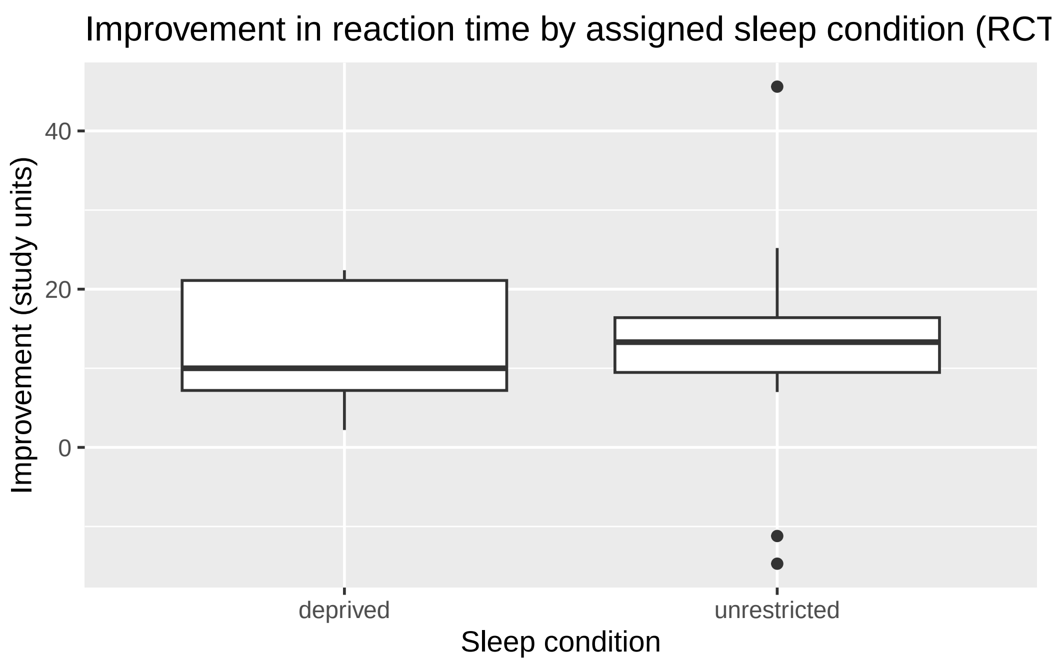
```r
sleep_rct |>
  group_by(sleepcondition) |>
```

```
  summarize(
    n = n(),
    mean_improvement = mean(improvement),
    sd_improvement = sd(improvement),
    median_improvement = median(improvement),
    iqr_improvement = IQR(improvement)
  )
```

```
# A tibble: 2 x 6
  sleepcondition     n mean_improvement sd_improvement median_improvement
  <fct>          <int>            <dbl>          <dbl>              <dbl>
1 deprived           9             12.4           7.61               10
2 unrestricted      12             12.4          15.5                13.3
# i 1 more variable: iqr_improvement <dbl>
```

Plots

```
ggplot(sleep_rct, aes(x = sleepcondition, y = improvement)) +
  geom_boxplot() +
  labs(
    title = "Improvement in reaction time by assigned sleep condition (RCT framing)",
    x = "Sleep condition",
    y = "Improvement (study units)"
  )
```



Welch two-sample t-test (Lecture 25 recap)

Conditions (practical): Independence between all students in both groups; small sample sizes so we'll have to assume the populations of improvement are approximately normal.

```
t.test(improvement ~ sleepcondition, data = sleep_rct, var.equal = FALSE)
```

```
	Welch Two Sample t-test

data:  improvement by sleepcondition
t = -0.0010785, df = 16.806, p-value = 0.9992
alternative hypothesis: true difference in means between group deprived and group unrestricted is not equal
95 percent confidence interval:
 -10.88355  10.87244
sample estimates:
    mean in group deprived mean in group unrestricted
              12.44444                   12.45000
```

Conclusion from RCT. If the design is truly randomized and well executed, a small p-value supports that the mean improvement differs by assigned sleep condition beyond what chance alone would typically produce.

Binary outcome in an RCT:

Story. 200 patients per arm; count **infections** (yes/no).

Hypotheses (population proportions). Let p_V and p_P be the **population** probabilities (or long-run proportions) of infection under **vaccine** and **placebo** assignment in the target setting.

- H_0: $p_V = p_P$ (no difference in infection probability between the two groups).
- H_a: $p_V \neq p_P$ (there is a difference in infection probability between the two groups).

```r
set.seed(2901)
vax_trial <- tibble(
  arm = factor(rep(c("Vaccine", "Placebo"), each = 200)),
  infected = c(rbinom(200, 1, 0.08), rbinom(200, 1, 0.14))
)

tab_v <- table(vax_trial$arm, vax_trial$infected)
tab_v
```

```
          0   1
  Placebo 174  26
  Vaccine 178  22
```

```r
infected_vaccine <- sum(vax_trial$infected[vax_trial$arm == "Vaccine"])
n_vaccine <- sum(vax_trial$arm == "Vaccine")
infected_placebo <- sum(vax_trial$infected[vax_trial$arm == "Placebo"])
n_placebo <- sum(vax_trial$arm == "Placebo")

prop.test(x = c(infected_vaccine, infected_placebo), n = c(n_vaccine, n_placebo), correct = FALSE)
```

```
	2-sample test for equality of proportions without continuity correction

data:  c(infected_vaccine, infected_placebo) out of c(n_vaccine, n_placebo)
X-squared = 0.37879, df = 1, p-value = 0.5383
alternative hypothesis: two.sided
```

```
95 percent confidence interval:
 -0.08366113  0.04366113
sample estimates:
prop 1 prop 2 
  0.11   0.13 
```

```
chisq.test(tab_v)$expected
```

```
         
            0  1
  Placebo 176 24
  Vaccine 176 24
```

```
chisq.test(tab_v)
```

```

	Pearson's Chi-squared test with Yates' continuity correction

data:  tab_v
X-squared = 0.21307, df = 1, p-value = 0.6444
```

Here **random assignment** makes a **low** infection rate in the vaccine arm **more** plausibly a **treatment effect** than the same table would be in an **observational** comparison.

When an RCT is impossible or unethical

Some groups cannot be randomly assigned (e.g. long-term **smoking** in teens).
